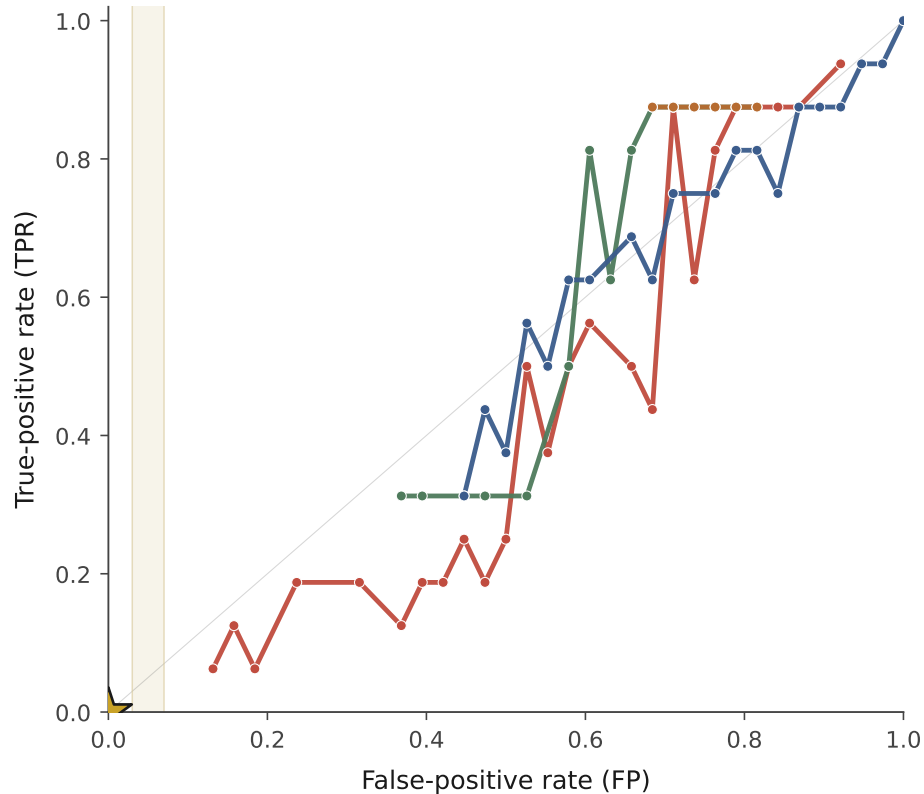
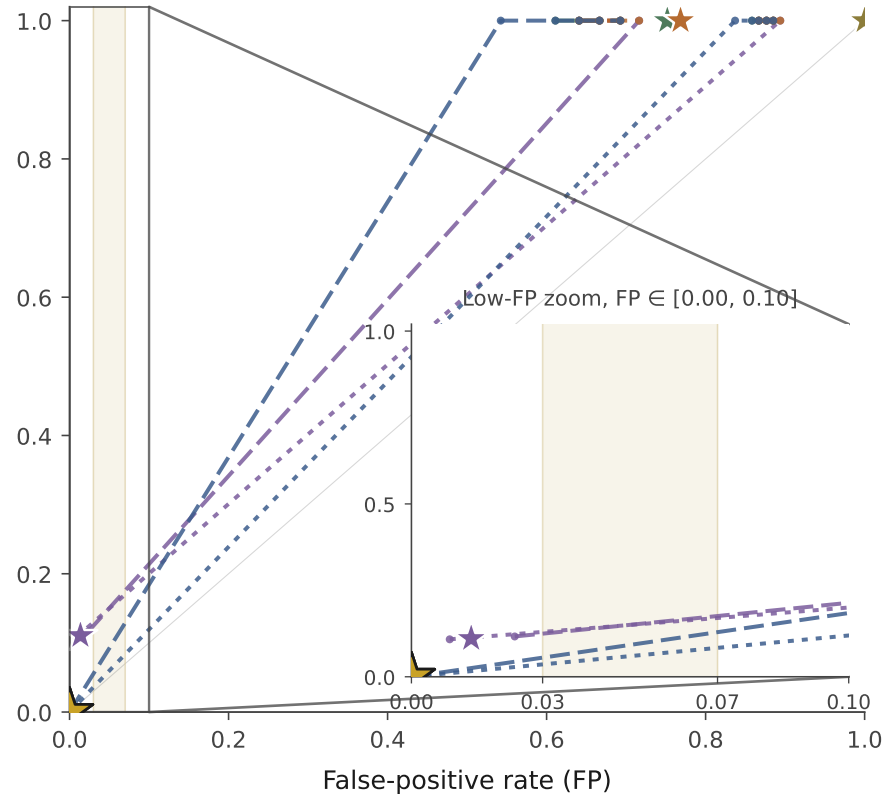


A. Markets — Volmageddon + COVID, late-30min window



B. Recommender — MovieLens-25M, horizons h=40/50/60



AC1 (recsys) CUSUM Page-Hinkley Variance EWS
 AC1-EWS (markets) Matrix Profile (recsys) Permutation Entropy (recsys)
 — h=40 (curve) ... h=60 (curve) ★ h=50 canonical (★, colored by family) Acceptance band [0.03, 0.07] ★ Loopzero pre-registered (★)